



Job design research and theory: Past, present and future [☆]

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ABSTRACT

This article reviews the research and theory that have focused on the design of jobs in organizations. We begin by summarizing some of the earliest work on this topic and then move to a discussion of several approaches to job design that attempted to address the shortcomings of this work. Next, we discuss several streams of contemporary research that have expanded the scope or deepened our understanding of job design. We conclude with a discussion of some future directions for research with an emphasis on job crafting, the effects of new work arrangements on the design of jobs, generational differences and reactions to job design, cultural differences and job design, and the impact of job design on organizational structures and employees' personal characteristics.

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1. Introduction

Over the past fifty years, few topics in the organizational sciences have attracted as much attention as job design (Clegg & Spencer, 2007; Fried, Levi, & Laurence, 2008; Hofmans, Gelens, & Theuns, 2014). The purpose of this article is to review the ideas, research and theory that have addressed this topic and to lay out several new directions for future research. We begin by defining job design and discussing the early work that was instrumental in shaping the direction of research on the topic. We then move to a discussion of the state of current research and theory on job design. Finally, we conclude with some ideas for future research.

2. Early work on job design

At its most basic level, job design refers to the actual structure of jobs that employees perform. Thus, job design focuses squarely on the work itself—on the tasks or activities that employees complete for their organizations on a daily basis. The earliest work on the topic of job design can be traced to the writings of Babbage (1835) and Smith (1850) who argued that if jobs were specialized and simplified to the greatest extent practicable, employees would be able to hone their job-related skills and devote their full

attention to very few tasks. These enhanced skills and focused attention were then expected to contribute to improved employee efficiency at work.

Job simplification and standardization were also critical parts of the scientific management philosophy developed by Taylor (1911). Taylor's basic idea was to design entire work systems with standardized operations and highly simplified jobs so that employees had little personal discretion at work and any unnecessary motions could be eliminated (Lawrence, 2010). Also, in scientific management there was little opportunity for employee involvement in the design process itself—management designed jobs and imposed these designs on employees in a top-down fashion.

Scientific management had a substantial impact on the job design practices of many firms. For example, in a study of manufacturing firms in the 1950s, researchers showed that most jobs were designed consistent with scientific management principles (Davis, Canter, & Hoffman, 1955). During that same period, research also began to show that many employees did not care much for the simplified jobs they were required to perform in scientific management—so much so that they often behaved in ways that negated the efficiencies that had been built into the work. Such counter-productive behaviors included tardiness and productivity restriction (Walker & Guest, 1952).

In an effort to deal with these counter-productive behaviors, a number of scholars developed approaches to job design that would allow employees to achieve high levels of performance without incurring the costs associated with simplified work (see Davis & Taylor, 1972). Many of these approaches were based on Herzberg's (1966) Motivation-Hygiene Theory which posited that

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in order to enhance employee performance and job satisfaction, jobs should be *enriched* rather than simplified. That is, work should be designed to include “motivators” that would foster employee responsibility, achievement, growth in competence, recognition, and advancement.

Herzberg's ideas were considered revolutionary at the time and spawned a great deal of research and a large number of successful job enrichment projects (see Herzberg, 1976; Paul, Robertson, & Herzberg, 1969). And as Herzberg expected, many of these projects demonstrated that boosting a job's standing on the motivators generally led to beneficial outcomes such as increases in employee work effectiveness and job satisfaction.

Unfortunately, despite its merits, there were several difficulties with Herzberg's approach. For example, his approach provided little guidance about the specific properties that might be introduced during job enrichment and did not offer a technology for measuring the presence of these job properties. Also, like scientific management, job changes were imposed by management in a top-down fashion without giving employees opportunities to suggest changes in the work itself (Paul et al., 1969). Finally, Herzberg did not allow for the possibility that there may be differences in how responsive employees were to job enrichment despite the results of early studies showing that some people may respond more positively than others to enriched work (see Davis & Taylor, 1972).

Research in the late 1960s and 1970s attempted to address these issues. One major study during this period was conducted by Turner and Lawrence (1965). These authors examined six “Requisite Task Attributes” (i.e., variety, autonomy, required interaction, optional interaction, knowledge and skill required, and responsibility) they expected to shape employee reactions and then created a summary measure (i.e., the RTA index) by formulating a linear combination of the six attributes. Results showed that the index was positively associated with the satisfaction and attendance of employees who worked in factories located in small towns. For employees in urban areas, however, the RTA index was negatively related to satisfaction and unrelated to attendance. Turner and Lawrence concluded that employees with different subcultural backgrounds reacted differently to high RTA jobs, and later research by Blood and Hulin (1967) supported these arguments.

Hackman and Lawler (1971) extended the Turner-Lawrence work by focusing on the effects of four job characteristics (i.e., autonomy, variety, task identity, and feedback) they expected to be strongly related to an employee's internal work motivation (i.e., the extent to which the employee feels good when performing well, and feels bad when performing poorly). In addition, these authors suggested that the previously found differences in how members of subcultural groups responded to their jobs might most simply be explained in terms of employees' needs for growth and development at work. Specifically, they argued that the stronger an employee's need for growth, the more likely he or she would be to respond positively to jobs high on the four characteristics.

Results of the Hackman-Lawler research provided support for most of their predictions—employees who worked on jobs high on the four characteristics were more internally motivated, satisfied with their jobs, and productive. Further, employees with high growth need strength (GNS) showed more positive responses to the four characteristics than those with lower GNS.

The Hackman and Lawler (1971) study stimulated a good deal of research and also provided the foundation for Job Characteristics Theory (JCT; Hackman & Oldham, 1976, 1980) which became the most widely-researched and debated approach to job design from the late 1970s until the present day (Ghosh, Rai, Chauhan, Gupta, & Singh, 2015; Grant, Fried, & Juillerat, 2010). JCT extended the Hackman-Lawler work in several ways. First, it focused on five

(versus four) core job characteristics that were expected to contribute to an employee's internal work motivation and other outcomes. These were: skill variety (i.e., the degree to which the job requires a variety of different activities involving the use of different skills), task identity (i.e., the degree to which the job requires doing a whole and identifiable piece of work), task significance (i.e., the degree to which the job has an impact on the lives of others), autonomy (i.e., the degree to which the job provides substantial freedom to the employee), and job-based feedback (i.e., the degree to which carrying out the work provides the employee with performance information). Second, JCT provided a testable theoretical framework that explained the effects of these job characteristics on employee outcomes (e.g., internal motivation, job satisfaction, performance). Each of the job characteristics was expected to contribute to the outcomes via its effects on one of three employee psychological states. Skill variety, task significance, and task identity were expected to contribute to the *experienced meaningfulness* of the work. Autonomy was expected to contribute to the *experienced responsibility* for work outcomes and feedback was expected to provide direct *knowledge of the results* of the work. Third, JCT posited that three conditions would moderate the effects of job characteristics. In addition to GNS, employees were predicted to respond most positively to the five characteristics when (a) they had job-relevant knowledge and skill and (b) were satisfied with the work context (i.e., with security, pay, supervision, coworkers). Employees satisfied with the context were expected to respond positively to jobs high on the characteristics because they were able to focus their attentions directly on the work itself, consistent with Herzberg's (1966) position about the role of the context and hygiene factors in job enrichment. Finally, the Job Diagnostic Survey (JDS) was created to assess jobholder descriptions of the job characteristics along with other constructs central to the theory.

Hundreds of studies tested JCT using cross-sectional and longitudinal designs, and the results provided some support for the theory's major propositions (Fried & Ferris, 1987; Humphrey, Nahrgang, & Morgeson, 2007). For example, there was strong support for the expected positive relations between the core characteristics and employee attitudinal outcomes (e.g., internal motivation, job satisfaction), however, the relations between the job characteristics and behavioral outcomes (e.g., attendance, performance) were relatively modest in magnitude (Humphrey et al., 2007; Koppelman, 1985). Also, results provided strong support for experienced meaningfulness as a mediator, but only weak support for experienced responsibility, and little support for knowledge of results (Humphrey et al., 2007). Results involving the proposed moderators provided only mixed support for GNS (Fried & Ferris, 1987; Lohr, Noe, Moeller, & Fitzgerald, 1985; Spector, 1985) and for context satisfactions (DeVaro, Li, & Brookshire, 2007; Fried et al., 2008; Ghosh et al., 2015), with some studies showing the expected positive relations between the job characteristics and outcomes for those high on GNS and context satisfactions, and others showing weak, statistically nonsignificant relations. No studies directly tested the moderating effect of knowledge and skill or the expected joint moderating effect of that variable, GNS and context satisfactions.

In summary, early work on the topic of job design focused on the effects of simplified, standardized jobs and demonstrated that employees often exhibited counter-productive behaviors in response to such jobs. Herzberg (1966) and other scholars addressed these issues and argued that employees would actually be more productive and satisfied if their jobs were enriched versus simplified. Later research extended and refined Herzberg's ideas and focused on several specific job properties (e.g., autonomy, feedback) expected to boost employees' motivation, satisfaction and work effectiveness, and the possibility that there were

individual differences in how employees responded to such properties. This latter work included research on JCT (Hackman & Oldham, 1980) which was tested in numerous studies that obtained generally mixed results.

3. Expanding and deepening our understanding of job design

The mixed and inconsistent results obtained in early studies testing JCT and other approaches served to stimulate interest in new areas of research that focused on enhancing our understanding of the design of jobs. Some of this newer work attempted to refine and extend JCT while other work focused on developing entirely new perspectives that addressed job design. Several of these research topics are described below.

3.1. The measurement of job characteristics

A number of investigations explored issues concerning the measurement of job characteristics. One set of studies examined the factor structure of the 15 JDS items designed to measure the five characteristics in JCT and reported solutions inconsistent with the a priori five-factor structure (e.g., Birnbaum, Farh, & Wong, 1986; Dunham, Aldag, & Brief, 1977). Research then examined whether the five negatively worded JDS items were responsible for the inconsistent solutions (Harvey, Billings, & Nilan, 1985; Idaszak & Drasgow, 1987). Although a few studies demonstrated that replacing the negatively worded items with positively worded items resulted in a better fit with the a priori structure (Idaszak, Bottom, & Drasgow, 1988; Idaszak & Drasgow, 1987), results of other research showed that the revised items did not improve upon the originals in terms of predicting employee outcomes (Cordery & Sevastos, 1993; Kulik, Oldham, & Langner, 1988). Likely as a result of these latter findings, little research now deals with dimensionality issues and most contemporary work examining job characteristics uses either the original JDS or alternatives such as the Job Characteristics Inventory (JCI; Sims, Szilagyi, & Keller, 1976) and the Work Design Questionnaire (WDQ; Morgeson & Humphrey, 2006).

A second set of investigations addressed the objectivity of employee ratings of their jobs. JCT argued that jobholders consider the objective properties of their jobs and then provide accurate descriptions of these properties on the JDS. Scholars questioned this argument (see Roberts & Glick, 1981) and three separate streams of research explored this issue. The first of these examined whether employee job ratings converged with ratings made by observers (e.g., researchers) who were expected to provide more accurate descriptions of the work (e.g., Algera, 1983; Birnbaum et al., 1986). Meta-analyses of these studies showed moderate convergence between employee and observer ratings (Fried & Ferris, 1987; Spector, 1992), suggesting that employees provided generally accurate descriptions of their jobs.

A second research stream, based on Social Information Processing Theory (Salancik & Pfeffer, 1978), proposed that employees based their ratings on social information about the job available in the environment and not on objective job features. Research tested this idea by contrasting the effects of objective job conditions with informational job cues provided by others (e.g., Griffin, 1983; O'Reilly & Caldwell, 1979). Meta-analyses of this work showed that information provided by others did have an impact on employee job ratings, but that the effect of this information was weaker than that of objective job properties (Spector, 1992; Taber & Taylor, 1990).

Finally, several studies focused on the possibility that employees' personal dispositions (e.g., core self-evaluations, personality) shaped their descriptions of job characteristics. The basic argument

was that employees describe their jobs as more complex and challenging in nature because they are predisposed to perceive all aspects of their work positively. A number of investigations were conducted that provided mixed support for this argument (e.g., Frew & Bruning, 1987; Judge, Bono, & Locke, 2000; Judge, Locke, Durham, & Kluger, 1998; Kittinger, Walker, Cope, & Wuensch, 2009; Spector, Fox, & Van Katwyk, 1999; Spector, Jex, & Chen, 1995; Thomas, Buboltz, & Winkelspecht, 2004; van den Berg & Feij, 1993, 2003). For example, a meta-analysis of research on core self-evaluations (i.e., self-esteem, self-efficacy, locus of control, neuroticism) showed that a composite measure of self-evaluations was positively related to a summary measure of job characteristics and to some individual job characteristics (e.g., autonomy) (Chang, Ferris, Johnson, Rosen, & Tan, 2012). However, other research showed generally weak relations between measures of personality and job characteristics (e.g., Spector et al., 1999; van den Berg & Feij, 1993). For example, van den Berg and Feij (2003) showed a significant relation between self-report measures of feedback and achievement motivation but nonsignificant relations between measures of extraversion and neuroticism and measures of feedback, autonomy, and variety.

In total, the results of research concerned with the objectivity of employee job ratings suggests that employees provide generally accurate descriptions of their job characteristics although these descriptions might be shaped somewhat by employees' personal dispositions and external conditions. As a result of these findings, little research now focuses on this topic and most contemporary studies in the area of job design use employees' self-reports of their job characteristics (e.g., Astrauskaite, Notelaers, Medisauskaitė, & Kern, 2015; Barrick, Thurgood, Smith, & Courtwright, 2015; Hofmans et al., 2014).

3.2. The search for new moderating variables

Possibly due to the generally weak results involving the moderating conditions included in JCT (i.e., GNS and context satisfactions), researchers began exploring variables that might effectively moderate relations between job characteristics and employees' personal and work outcomes. Many of these studies were successful in identifying variables that had a significant impact on the relations between job characteristics and outcomes. For example, Grant et al. (2007) showed that beneficiary contact moderated task significance-persistence relations. Johns, Xie, and Fang (1992) showed that relations between a summary measure of job characteristics and employee outcomes were stronger when employees perceived that their pay was not contingent on performance—that is, when pay was not considered a mechanism to control their performance (Ryan & Deci, 2002). And Bond, Flaxman, and Bunce (2008) demonstrated that psychological flexibility moderated relations between autonomy and employee stress and attendance. In the paragraphs below, we discuss six groups of moderating variables that have received the most attention in the literature from the 1970s to the present day.

3.2.1. Personality

One set of studies examined the possibility that individuals react to job characteristics as a function of their standing on various personality traits, including the Big 5 (i.e., openness to experience, neuroticism, conscientiousness, extraversion, agreeableness) (Barrick & Mount, 1993). That is, rather than arguing employee personality shapes job perceptions as discussed above, these studies argued that individuals reacted differently to jobs as a function of their personality. Results of a few investigations provided some support for the argument. For example, Grant (2008) found that an intervention designed to boost task significance had a stronger effect on performance when employees

scored low on conscientiousness. Mount, Barrick, and Stewart (1998) showed that employees holding jobs that required interpersonal interaction (e.g., customer service) performed best when they scored high on conscientiousness, agreeableness and emotional stability. Fortunato and Stone-Romero (2001) showed that positive affectivity moderated overall job complexity-job satisfaction relations. Raja and Johns (2010) examined the effects of the Big 5 on relations between a summary job index and three outcomes (i.e., organizational citizenship behaviors, creativity, performance). Results showed that neuroticism was the most consistent moderator with neurotics responding poorly to jobs high on the core characteristics. Finally, and in contrast to other studies, Thomas et al. (2004) demonstrated that several personality measures (e.g., extraversion) had nonsignificant effects on relations between employee job characteristics and their job satisfaction.

In total, the research described above suggests that employees generally respond differently to job properties as a function of their personalities. Research is now needed to extend this work perhaps by identifying the specific personality traits that are especially relevant for each of a job's core characteristics.

3.2.2. Career stage

A second set of studies examined whether employees responded differently to job characteristics as a function of their career stage (see Fried, Grant, Levi, Hadani, & Slowik, 2007). This work posited that individuals at the early and late stages (10+ years) of their careers would be unresponsive to core job characteristics. Those at the early stages were expected to be preoccupied with becoming accepted in the work unit; those at the late stages were expected to find complex jobs non-stimulating and customary. Only those in the middle stages were expected to be responsive. Two studies by Katz (1978a, 1978b) provided some support for these arguments showing stronger job design-satisfaction relations during middle career stages than during early and late stages. However, other studies (e.g., Johns et al., 1992; Kemp & Cook, 1983) failed to replicate the results obtained by Katz. Likely as a result of these mixed findings, little contemporary research focuses on the moderating effects of career stages. Future research might attempt to extend this work by examining the specific job characteristics (e.g., autonomy) that have especially strong effects on outcomes at different stages of employees' careers.

3.2.3. Physical context

Research in the 1980s and 1990s examined the possibility that elements of the physical context (e.g., boundaries, openness, density) moderate relations between job characteristics and outcomes (Fried et al., 2008). The basic argument was that the presence of physical elements that limit unwanted exposure to others enhance individuals' ability to concentrate at work which should be especially beneficial when jobs are complex and challenging. Results of these studies were generally inconsistent with some providing findings that supported the argument (Block & Stokes, 1989), and others reporting results that offered little support (e.g., Oldham, Kulik, & Stepina, 1991; Sundstrom, Burt, & Kamp, 1980; Sutton & Rafaeli, 1987). Research is now needed to extend and refine this research perhaps by examining whether there are specific features of the physical context (e.g., openness or flexible workspaces) that enable employees with certain job characteristics (e.g., required interaction) to most effectively complete their work.

3.2.4. Others' jobs

Some research focused on the possibility that the jobs of others (e.g., employees in the work unit) influenced the way employees responded to their own jobs. Based on equity theory (Adams, 1963), individuals were expected to respond most positively when their jobs were comparable in complexity to the jobs of others.

Results of a few investigations provided some support for this argument (e.g., Montagno, 1985; Oldham, Kulik, Ambrose, Stepina, & Brand, 1986; Oldham et al., 1982). However, the ideas about the importance of the jobs of others did not capture the attention of later researchers, and little research from the 1990s to the present day has focused on this topic.

3.2.5. Uncertainty

Several investigations examined the effects of uncertainty (i.e., lack of predictability in tasks and requirements; Wall, Cordery, & Clegg, 2002) on the relations between job characteristics and outcomes. The argument was that at high levels of uncertainty, increasing job control should enhance performance because it provides an opportunity for employees to learn about tasks (Wall et al., 2002). By contrast, when uncertainty is low, increasing job control should have little effect since there are few problems that require attention and little need for learning. Previous studies have provided general support for these arguments (e.g., Leach et al., 2013; Wall, Corbett, Martin, Clegg, & Jackson, 1990; Wright & Cordery, 1999) and more research is now needed to examine whether uncertainty interacts with several job characteristics to affect employee performance and well-being.

3.2.6. Job Demands-Resources (JD-R) framework

This framework described by Demerouti, Bakker, Nachreiner, and Schaufeli (2001) (which has its origins in Job Demand-Control model; Karasek, 1979) suggests that elements of the organizational context interact with the design of jobs. According to the JD-R framework, occupational factors can be classified into two categories: job demands and job resources. Demands include elements that require physical or psychological effort; job resources (e.g., job characteristics) help the employee deal with job demands. Specifically, the framework suggests that high levels of some job characteristics allow employees to cope with a demanding environment resulting in more positive outcomes. Previous research provides some support for these arguments (see Bakker & Demerouti, 2007). For example, Bakker, Demerouti, and Euwema (2005) showed that employees who experienced high job demands reported lower burnout rates if their jobs were autonomous and they received feedback. Unfortunately, it is not yet clear what demands are most significant and what features of the work meet these demands. More research is now needed to address these issues.

In summary, the generally weak results involving the JCT moderators may have stimulated research that focused on identifying personal and contextual conditions that were effective in moderating relations between job characteristics and employee outcomes. Much of this work showed considerable promise (e.g., studies of personality, task uncertainty, and job demands) and research is now needed to extend these findings and to integrate the moderators into a comprehensive model of job design. In addition, future work might continue to search for other conditions (e.g., specific types of financial reward programs) that moderate the effects of job design thereby helping us to better understand the conditions under which job characteristics enhance employee outcomes.

3.3. Effects of the organizational context on the design of jobs

Research in the late 1970s and 1980s examined the possibility that elements of the organizational context (e.g., structure, technology) shape the characteristics of employees' jobs (Pierce, 1979; Rousseau, 1978b). The general argument was that employee job characteristics differed as a function of various contextual elements and several studies provided support for this position. For example, Rousseau (1977) showed that employee descriptions of the core characteristics varied as a function of Thompson's

(1967) technology categories: long-linked, mediating, and intensive. Mean levels of the job characteristics were found to be lowest in organizations with long-linked technologies and highest in those with mediating. Other studies examined relations between job characteristics and the structural properties (e.g., size, centralization, formalization, number of levels) of the organization (Oldham & Hackman, 1981) and the department (Rousseau, 1978a). Results of this research showed that many of the structural properties were substantially related to the job characteristics. For example, Pierce and Dunham (1978) found that departmental formalization and centralization were significantly and negatively associated with employee descriptions of the core characteristics, suggesting that centralized, hierarchical organizations constrain the emergence of these job dimensions. Finally, some researchers argued that the work unit's physical space could directly shape the characteristics of jobs rather than serve as a moderator as discussed earlier. The argument was that configurations that encouraged unwanted interpersonal intrusions (e.g., few barriers) could increase monitoring and disruptions at work which, in turn, affected employee job descriptions. A number of studies provided support for this general position (e.g., Oldham & Brass, 1979; Oldham & Rotchford, 1983; Schuler, Ritzman, & Davis, 1981). For example, Szilagyi and Holland (1980) showed that employees who moved from a low- to a high-density office reported lower levels of autonomy.

In total, the research described above demonstrated that many elements of the organizational context can shape employee job descriptions. Given these significant findings, little contemporary research focuses on the effects of these elements of the context on the design of jobs. However, we believe that research on a related topic—the effects of *occupational* contexts on employee jobs (see Dierdorff & Morgeson, 2013)—does warrant additional attention as does research on the effects of new work arrangements on jobs, which we discuss later in this article.

3.4. An interdisciplinary perspective of job design

Campion and his colleagues (Campion, 1988; Campion & McClelland, 1991, 1993) broadened the discussion of research on job design by proposing a framework that integrated motivational approaches (e.g., JCT) with three alternatives: perceptual/motor, mechanistic, and biological. An instrument (i.e., Multimethod Job Design Questionnaire; Campion & Thayer, 1985) was also developed to assess the elements associated with each approach. Campion argued that each of the approaches had both benefits and costs and that it was critical to consider these before undertaking any type of job design intervention. For example, designing jobs from a motivational perspective might enhance employee effectiveness but more training may be required.

Research provided some support for the integrative framework. Specifically, the outcomes that were expected to be connected to each individual approach typically were while each approach was generally only weakly related to the outcomes that did not correspond to it (Campion, 1988). Given these findings, little current research tests the entire integrative framework. However, a study by Morgeson and Campion (2002) extended the early work by showing that there were ways to redesign jobs and minimize the tradeoffs between the motivational and mechanistic approaches such that both satisfaction and efficiency were enhanced. Future research might extend this work by examining strategies that minimize tradeoffs involving all four of the approaches to job design.

In summary, the inconsistent findings obtained in early studies of JCT and other approaches to job design stimulated research on a number of new topics. These included attempts to improve the measurement of job characteristics, to identify personal and

contextual conditions (e.g., personality and uncertainty) that were effective in moderating relations between job characteristics and employee outcomes, and to develop frameworks (e.g., the interdisciplinary perspective) that addressed the design of jobs. Much of this work contributed substantially to the field and has provided a direction for contemporary research on job design, discussed in the next section.

4. Current research on job design

Thus far we have focused on early research and theory on the topic of job design and several later research streams that deepened our knowledge of the topic. But research on the design of jobs has not stopped. Many studies continue to examine how to best design jobs to enhance employees' well-being and work effectiveness. Below we discuss some current streams of research that attempt to broaden the discussion of job design and that have received considerable research attention.

4.1. Alternative job characteristics

Interest in job characteristics beyond those suggested by JCT and Campion's framework has grown stronger in recent years possibly as a result of the global shift from a manufacturing economy to a knowledge and service economy (Grant & Parker, 2009). Many authors now acknowledge that the job characteristics that received much attention in the past may not capture all the dimensions of jobs that shape employee responses in contemporary organizations (see Grant & Parker, 2009; Oldham & Hackman, 2010).

Among the job characteristics receiving the most attention in the current literature are those involving the social dimensions of the work (Grant & Parker, 2009). Several measures developed in the 1960s and 1970s were designed to tap social job dimensions but received little attention in those early years of job design research. These dimensions might be especially relevant for work in the service sector and include measures of "optional interaction" and "required interaction" (Turner & Lawrence, 1965), "interaction with others" (Sims et al., 1976), "friendship opportunities" (Hackman & Lawler, 1971), and "feedback from agents" (i.e., feedback from coworkers or supervisors) (Hackman & Oldham, 1975). In addition, the WDQ included three other social dimensions: "interdependence," "interaction with others outside the organization," and "social support" (i.e., the extent to which a job provides opportunities for assistance from others).

Contemporary researchers have suggested a number of other job characteristics that might contribute to employee responses beyond the characteristics included in JCT and the social characteristics described above (e.g., Parker, Wall, & Cordery, 2001). For example, Humphrey et al. (2007) suggested three groups of job characteristics: motivational, contextual, and social. The first category includes the JCT dimensions plus four others: information processing, job complexity (i.e., the extent to which the job is multifaceted), specialization, and problem-solving (i.e., the extent to which the job requires the production of unique solutions). The second category includes three dimensions related to the biological approach suggested by Campion and McClelland (1991): physical demands, work conditions (i.e., the extent to which there are temperature or noise issues on the job), and ergonomics (i.e., the extent to which work permits appropriate movement). The final category includes four social dimensions described earlier: interaction outside, social support, interdependence, and feedback from agents.

The Humphrey et al. (2007) meta-analysis included studies that examined all three categories above and showed that among the work context characteristics, physical demands was negatively

related to job satisfaction while working conditions was positively related to it. As noted earlier, the JCT motivational characteristics had generally positive relations to employee reactions. Moreover, two of the motivational dimensions not included in JCT (information processing and job complexity) showed positive relations to job satisfaction. Results also showed that all four of the social characteristics were positively related to employee job satisfaction, although only feedback from agents and interdependence were related to performance ratings. Finally, the full set of social characteristics explained variance in job satisfaction and performance beyond that explained by the set of motivational characteristics.

In total, these results underscore the value of examining job characteristics beyond those included in JCT and the integrative framework—especially those involving social dimensions. Research is now needed to determine whether the social characteristics are especially relevant for jobs in the service sector and are more strongly related to the responses of employees who work in that sector. Work is also needed to identify the specific social dimensions that are most relevant to employee motivation. Finally, future research might examine the connections between all categories of job dimensions and outcomes that have received relatively little attention in the literature. We now turn to a discussion of several such outcomes.

4.2. Alternative outcomes

Much of the early research on job design focused on the effects of jobs on traditional outcomes (e.g., satisfaction, performance). But in recent years, research has extended this work and examined the possibility that the way jobs are designed might have a broader impact on individuals and their employing organizations. For example, recent research has focused on the effects of jobs on such outcomes as bullying (Astrauskaite et al., 2015; Baillien, De Cuyper, & De Witte, 2011), private lives (Schuller, Roesler, & Rau, 2014), and volunteering (Pajo & Lee, 2011). Several additional outcomes have been addressed more comprehensively and we discuss them below.

4.2.1. Creativity

This refers to the production of ideas concerning organizational products, practices, services, or procedures that are (a) novel and (b) potentially useful to the organization (Amabile, 1996). Commentators have long argued that employee creativity is of value because it provides management with ideas that might be implemented and introduced in the organization as a whole or in the marketplace (see Oldham & Baer, 2012).

Several studies have examined relations between the motivational characteristics of employees' jobs and their creativity. In general, results show that individuals exhibit high creativity when they work on jobs that include high levels of characteristics such as autonomy, skill variety, and job-based feedback (e.g., Coelho & Augusto, 2010; Farmer, Tierney, & Kung-McIntyre, 2003; Oldham & Cummings, 1996; Shalley, Gilson, & Blum, 2009; Zhang & Bartol, 2010). Less is known about the effects of the social and context job dimensions described in the paragraphs above. Research is now needed to investigate the effects of these characteristics on employee creativity. For example, studies might examine whether individuals who work interdependently or with others outside the boundaries of the work unit acquire new ideas or stimulation from those others that then enhance their own creative idea development.

4.2.2. Organizational citizenship behaviors (OCBs)

These refer to discretionary behaviors that an employee exhibits that are not required by the employee's role or job description (Organ, 1988). Podsakoff, MacKenzie, Paine, and Bacharach (2000)

described several OCBs including, altruism, courtesy, sportsmanship, conscientiousness, and civic virtue. A meta-analysis of the literature showed that three "task characteristics" (i.e., task feedback, task routinization, and intrinsically satisfying tasks) were significantly related to these OCBs (Podsakoff, MacKenzie, & Bommer, 1996). Other research examined relations between motivational job characteristics and summary indices composed of scores on several OCB dimensions (e.g., Chen & Chiu, 2009; Piccolo & Colquitt, 2006; Purvanova, Bono, & Dzieweczynski, 2006). In general, consistent with results reported in the Podsakoff et al. (1996) meta-analysis, results of these studies showed positive relations between job characteristics (e.g., autonomy, skill variety, task significance) and the OCB indices (e.g., Chiu & Chen, 2005; Purvanova et al., 2006). Research is now needed to investigate the effects of the social and context job dimensions discussed earlier and to clarify which of the categories of job characteristics most strongly relate to the various OCBs.

4.2.3. Employee health and well-being

Another research stream examines whether the work itself has unintended effects on psychological (e.g., stress) and physical (e.g., obesity) outcomes related to employee well-being (e.g., Bond et al., 2008; Frew & Bruning, 1987; Lambert, Hogan, Dial, Jiang, & Khondaker, 2012). Results of previous research suggest that several job characteristics often relate substantially to well-being indicators—especially psychological outcomes. For example, the Humphrey et al. (2007) meta-analysis showed that job characteristics (e.g., working conditions, autonomy, task identity, feedback from agents) were related negatively and in a linear fashion to the outcomes of stress and exhaustion. These results suggest that working under poor conditions and on jobs low on the motivational characteristics can adversely impact employee mental health.

Results of research focusing on physical outcomes are less clear. Some studies in this set do suggest a linear relation between the motivational job characteristics and physical outcomes, consistent with results for stress and exhaustion. For example, Igic, Ryser, and Elfering (2013) showed that the presence of autonomy and personal control reduced employees' spinal shrinkage which could then lower the incidence of back pain. But studies focusing on other physical outcomes suggest more complex relationships. For example, Melamed, Fried, and Froom (2001) showed a significant relation between an overall measure of job complexity and systolic and diastolic blood pressure rates only for employees exposed to high levels of noise. The design of jobs had little impact on blood pressure when noise levels were low. Fried et al. (2013) showed that an index of the core characteristics had a curvilinear relation to two indicators of employee obesity (i.e., waist circumference and waist-hip ratio) such that the lowest rates were present when employee jobs were moderately complex. Higher rates of obesity were present when jobs were either simple or complex, suggesting that employees find jobs at these levels as either under- or over-stimulating.

Research is now needed that focuses on the connections between employee psychological and physical health and the dimensions of jobs within the several categories described earlier. Such studies might attempt to sort out which dimensions are relevant for which of the health outcomes and the circumstances under which these dimensions have their strongest effects. In particular, given the findings of Fried et al. (2013) and those of studies focusing on traditional outcomes (e.g., Xie & Johns, 1995), research is needed that examines the curvilinear effects of motivational, social and contextual characteristics on indicators of employee well-being. In addition, research is needed on the effects of beneficiary contact on employee well-being. Previous research has shown that contact with beneficiaries can boost the positive effects

of task significance on employee motivation (Grant et al., 2007). However, other studies suggest that such contact can be stressful and result in burnout as it often requires employees to deal with individuals who engage in unfriendly behaviors or verbal aggression (Grandey & Diamond, 2009; Grandey, Dickter, & Sin, 2004). Research is now needed that examines the circumstances under which contact with beneficiaries has positive effects on employees and those under which it adversely affects employee well-being (Grant & Parker, 2009).

4.3. Mediators of job characteristics

Possibly due to the weak results discussed earlier involving the three mediators included in JCT, a number of studies have examined conditions that might effectively mediate the effects of job characteristics on a variety of employee outcomes. One attempt to deal with these earlier results focused on the one psychological state in JCT that was shown to be an effective mediator (Johns et al., 1992): experienced meaningfulness. The Humphrey et al. (2007) meta-analysis examined this mediator across a number of motivational characteristics and traditional outcomes and showed that it was quite effective in explaining the relationships.

Several other conditions have been suggested as possible mediators of the relationships between job characteristics and outcomes. For example, Parker et al. (2001) proposed that high levels of the motivational characteristics might enable employees to make “quick responses” to everyday disruptive events which could then boost their performance. A number of authors (e.g., Clegg & Spencer, 2007; Parker, 1998; Parker et al., 2001) argued that job characteristics allow individuals to develop their self-efficacy which, in turn, enhances their personal and work outcomes. Other authors have suggested psychological ownership (Pierce, Jussila, & Cummings, 2009), enjoyment (Hofmans et al. (2014), and empowerment (Liden, Wayne, & Sparrowe, 2000) as mediators. And Grant (2008) proposed perceived “social impact” and “social worth” as mediators between task significance and the outcomes of helping and dedication.

In addition to the mediators above, two recent studies focused on work engagement (i.e., high levels of personal investment in the work tasks performed on a job; Christian, Garza, & Slaughter, 2011) and obtained promising results. Shantz, Alfes, Truss, and Soane (2013) demonstrated that work engagement was effective in mediating the effects of the core characteristics on employee performance and an OCB index. A meta-analysis by Marinova, Peng, Lorinkova, Van Dyne, and Chiaburu (2015) showed that engagement effectively mediated the relations between three job characteristics (autonomy, task significance, and job complexity) and an index of employees’ “change-oriented behaviors” (e.g., taking charge, personal initiative, voice, creativity).

Research is now needed to compare and contrast the traditional (e.g., experienced meaningfulness) and alternative mediators discussed in the paragraphs above. Ideally, this research should examine a wide range of motivational, social and context job characteristics along with both traditional and non-traditional outcomes (e.g., performance, health, OCB’s) and attempt to determine which mediator(s) are most effective and for which outcomes.

4.4. The design of jobs for teams

Most of the early research on job design focused on strategies for designing the jobs of individuals who worked relatively independently of one another. But research in the 1980s began considering ways to design jobs when the work was to be performed by teams of employees. One model of team design that received a good deal of attention in the literature was proposed by Hackman (1987). He argued that the effectiveness of teams was a

joint function of three criteria: the level of effort applied to the team task, the knowledge and skill of team members, and the use of appropriate performance strategies. The first of the criteria was expected to be affected by the characteristics of the team task (e.g., the extent to which the team was autonomous, the team members used a variety of skills, and received trustworthy feedback). The second and third criteria were expected to be affected by the team composition and norms, respectively.

Little research tested Hackman’s full model, however several studies did examine the effects of one of the model’s features: the extent to which the team was autonomous and self-directed (e.g., Cordery, Mueller, & Smith, 1991; Glassop, 2002). While the majority of these studies reported that autonomous teams had positive effects on productivity and team member job satisfaction, a few investigations indicated that such teams had little impact on these outcomes (see Cohen & Bailey, 1997; Delarue, Van Hootegeem, Procter, & Burridge, 2008; DeVaro, 2008).

Given these inconsistent results, a number of recent investigations have attempted to determine if there are specific circumstances under which autonomous teams have positive effects. This research has contributed substantially to the literature by showing that autonomous groups are most effective when the team task is interdependent (Langfred, 2005), individual members have little autonomy (Drach-Zahavy, 2004), team members are not isolated from external influences (Haas, 2010), and when organizational reward, feedback and information systems are generally ineffective (Morgeson, Johnson, Campion, Medsker, & Mumford, 2006). Other studies have demonstrated that self-directed teams perform better with distributed-coordinated leadership (i.e., at least two leaders emerged in the team and identified one another as leaders) than with either centralized leadership or with no leaders (McIntyre & Foti, 2013; Mehra, Smith, Dixon, & Robertson, 2006). Finally, research by Behfar, Peterson, Mannix, and Trochim (2008) showed that autonomous teams produced more positive results when there was concern with both the task and integrating the interests of members and used this approach when managing conflicts within the team.

In addition to research focusing on the circumstances under which autonomous teams are most effective, recent research has examined potential unintended consequences of such teams (see Johns, 2010). One such study was conducted by Ingvaldsen and Rolfsen (2012) and showed that even if autonomous teams achieve high levels of intra-team performance, such teams may have difficulty coordinating their activities with other teams because of high team ownership experienced by team members. These results suggest that although the introduction of autonomous teams might boost intra-team performance when some of the conditions described above are in place, multi-team performance may actually suffer when these teams are present in organizations.

More research is now needed that investigates the specific circumstances under which autonomous teams enhance intra-team effectiveness without sacrificing inter-team coordination. In addition, research is needed that tests Hackman’s (1987) full model and attempts to integrate all of the conditions that are expected to shape the three criteria he proposed.

In summary, as discussed in the section above, research on the design of jobs continues to this day and focuses on such topics as alternative job characteristics, alternative outcomes, and new mediators of the relations between job characteristics and outcomes. This work has advanced our thinking about job design but many questions remain unanswered. It is to a discussion of some of these questions that we turn in the next section.

5. New research directions

Thus far we have reviewed some of the earliest work on the topic of job design and some of the research streams that refined

this work. We then focused on contemporary research and suggested several ways that work might be extended and developed. In this final section we discuss a number of new areas of research that we believe warrant attention.

5.1. Job crafting

Some of the most interesting research on job design to emerge in the early 2000s focused on the practice of job crafting (Wrzesniewski & Dutton, 2001). In general, job crafting involves employees taking the initiative in customizing their own work to fit their needs, values, skills and abilities (Bakker, Tims, & Derks, 2012; Nielsen & Abildgaard, 2012). This emphasis on employees shaping their own jobs is a major departure from early approaches to job design discussed in the pages above. In most of the early approaches, management was responsible for designing jobs and then imposed these designs on employees. JCT was somewhat different in that employees described their jobs on the JDS and then provided this information to management. But again, management was responsible for introducing all job changes after receiving input from employees.

The early work on job crafting was concerned with describing the ways that employees might shape their jobs. Wrzesniewski and Dutton (2001) suggested three general approaches: cognitive (changing the way the individual perceives his or her job), task (changing the content of work), and relational (changing the quality and amount of interaction with others). There have been a few attempts to refine these categories and to develop operational measures of crafting. Notable among these is work by Tims, Bakker, and Derks (2012) that describes four types of crafting activities based on the JD-R framework and provides measures of each. They are: increasing social job resources (maximizing feedback from the social context), increasing structural job resources (developing personal capabilities), increasing challenging job demands (taking on challenging new projects), and decreasing hindering job demands (organizing work such that it is the least stressful).

Several studies established that individual employees use one or more of the crafting activities discussed above (see Bakker et al., 2012; Petrou, Demerouti, Peeters, Schaufeli, & Hetland, 2012). And a few studies have demonstrated that these and other crafting activities are also used by groups of employees. For example, Leana, Appelbaum, and Shevchuk (2009) and others (e.g., Chen, Yen, & Tsai, 2014; Tims, Bakker, Derks, & Van Rhenen, 2013) examined “collaborative crafting” and showed that informal group members jointly determine how to shape the work in order to meet their shared objectives.

In addition to demonstrating that employees actively craft their jobs, other research has examined employees’ behavioral and attitudinal responses to these crafting activities. Although several authors have argued that job crafting activities should enhance employees’ responses because these activities improve person-job fit (see Chen et al., 2014; Lu, Wang, Lu, Du, & Bakker, 2014; Tims, Derks, & Bakker, 2016), research to date provides only mixed support for this argument (see Demerouti, Bakker, & Halbesleben, 2015; Tims, Bakker, & Derks, 2015; Tims et al., 2012). For example, for individual job crafting, Leana et al. (2009) showed that a general measure of crafting had a nonsignificant relation to the crafter’s performance. Bakker et al. (2012) showed that a job crafter’s performance was positively related to “increasing structural job resources” and “increasing challenging job demands” but unrelated to “increasing social resources.” Demerouti, Bakker, and Gevers (2015) showed that three dimensions of crafting (i.e., seeking resources, seeking challenges, and reducing demands) had nonsignificant relations to employee creativity. For collaborative crafting, Leana et al. (2009) showed that a general measure had a

positive relation to performance. But Tims, Bakker, Derks, and Van Rhenen (2013) showed that performance was positively related to “job resources and challenging job demands” and negatively related to “decreasing hindering demands.”

Results for attitudinal outcomes are also mixed (Leana et al., 2009; Tims, Bakker, & Derks, 2013; Tims et al., 2012). For individual crafting, Leana et al. (2009) showed that a general measure of crafting had a negative relation to job satisfaction and nonsignificant relations to organizational commitment and turnover intentions. van Hooff and van Hooff (2014) showed that “structural resources” and “challenging job demands” were negatively related to boredom but that “social resources” was unrelated to this outcome. For collaborative crafting, Tims, Bakker, Derks, and Van Rhenen (2013) showed that “job resources and challenging job demands” was positively related to engagement but that “hindering job demands” was unrelated to it. And Leana et al. (2009) showed that collaborative crafting was positively related to organizational commitment but had nonsignificant relations to satisfaction and turnover intentions.

In summary, a number investigations have shown that various job crafting activities have generally mixed relations to the job crafter’s behavioral and attitudinal outcomes. One explanation for these mixed results involves the crafting activities that have been examined in earlier studies. As noted earlier, most of the early work has focused on crafting activities suggested by the JD-R framework. But it may be that other crafting activities not suggested by this framework produce more positive, consistent results. For example, it may be that activities derived from JCT (e.g., attempting to boost significance by engaging in activities to connect with the immediate beneficiary of one’s work, reviewing one’s work before forwarding it to others as a way to boost job-based feedback, etc.) might have stronger effects than the activities examined in earlier studies. Research is now needed to develop alternative measures of crafting with strong psychometric properties and then to compare and contrast the effects of these measures with those investigated in earlier research. In addition, work is needed that examines whether the effects of crafting activities depend on critical boundary conditions (e.g., structural or social variables). It is possible that the mixed findings obtained in earlier studies were a result of such conditions—that is, some activities may have positive effects in some circumstances and weak effects in others. Work is now needed to develop a theoretical framework that describes these conditions and then examines their effects on individuals’ responses to various crafting activities. Also, research is needed that examines the possibility that individual and collaborative crafting activities combine to shape outcomes. For example, it may be that employees who engage in both individual and collaborative crafting exhibit the most positive responses at work. Finally, the early work on job crafting typically used employee self-reports of their activities. Research is needed to determine if these reports converge with others’ (e.g., supervisors, coworkers, researchers) ratings of the crafting activities of individuals and groups.

In addition to examining the direct effects of crafting activities on employee outcomes, a few studies have examined the conditions that mediate the effects of crafting activities (see Demerouti et al., 2015). For example, Tims, Bakker, Derks, and Van Rhenen (2013) demonstrated that work engagement mediated the effects of individual and collaborative crafting on performance. But research is now needed to compare and contrast several other mediating conditions, such as those presented in our earlier discussion of job characteristics. For example, studies might examine if crafting allows individuals to make quicker responses to changing events (Parker et al., 2001) and if crafting boosts employees’ experience of empowerment (Liden et al., 2000) or of meaningfulness (Hackman & Oldham, 1976). Any or all of these conditions might

explain the effects of crafting on job crafters' responses and are deserving of research attention.

Studies are also needed that examine the antecedents of job crafting. For example, [Vough and Parker \(2008\)](#) argued that increases in self-efficacy may lead employees to engage in crafting and a study by [Tims, Bakker, and Derks \(2014\)](#) provided support for this argument. [Bakker et al. \(2012\)](#) showed that individuals with proactive personalities ([Crant, 2000](#)) were most likely to engage in several types of crafting. [Clegg and Spencer \(2007\)](#) proposed that high employee performance might increase the supervisor's evaluation of his or her competence, which could then increase the supervisor's trust in the employee resulting in greater latitude to engage in crafting. They also suggested that employees are more likely to craft jobs with high autonomy and with job outcomes that are difficult to measure. Consistent with these latter arguments, studies have shown that an employee's discretion and autonomy at work relate positively to both individual and collaborative crafting (e.g., [Leana et al., 2009](#)). The Leana et al. study also showed that interdependence and supportive supervision were related to collaborative crafting, but not to individual crafting.

More work is now needed on the personal and situational conditions that encourage employees to engage in job crafting. For example, research might examine if individuals with strong needs for growth and development are likely to introduce new challenges into their own work in order to satisfy those needs ([Wrzesniewski & Dutton, 2001](#)) and if those with strong social needs are likely to introduce more social aspects into their jobs. Other studies might attempt to identify the characteristics of individuals who craft jobs not simply to fulfill their own needs but to benefit all others in the work unit. Research is also needed to determine whether an employee's effective performance increases his or her crafting activities as suggested by [Clegg and Spencer \(2007\)](#). If this is the case, studies might examine the specific crafting activities that are used and if these change as a result of increases in the supervisor's trust in the employee ([Fried, Levi, & Laurence, 2007](#)). If employee performance increases fail to boost his or her crafting activities, studies might then try to sort out the reasons for these results including examining the possibility that supervisors restrict or limit crafting because they perceive all such self-initiated activities as a threat to their status. Finally, research is needed to examine the possibility that successfully using one crafting activity (e.g., increasing structural job resources) might encourage individuals to use additional crafting activities when customizing their jobs.

Research might also examine the effects of various organizational practices and programs on individuals' crafting activities. It may be that individuals engage in crafting when the organization's reward system is designed to encourage employees' proactive behaviors. Also, if organizations use "crowdsourcing" ([Bayus, 2013](#)) to propose a problem to employees, this might encourage them to craft their jobs in order to develop the knowledge and skills necessary to address the problem (see [Afuah & Tucci, 2012](#)). Finally, future studies might examine the effect of the external economic environment on job crafting. It may be that in unstable economic conditions, individuals are most likely to engage in crafting as a mechanism to enhance their skills and marketability.

Our discussion thus far has focused on relations between various crafting activities and the responses of the job crafter. Little previous work has examined the responses of others in the work unit or broader organization to individual or collaborative job crafting. Since few employees in contemporary organizations work in complete isolation of others, it is quite possible that crafting activities on the part of a single individual or group have implications for the work and well-being of these others. For example, if job customization results in a product or service that is even slightly different from what existed before the customization, disruptions might occur down the line as other employees struggle to

deal with the modified service or product ([Oldham & Hackman, 2010](#)).

Research is now needed that examines the effects of various types of individual and collaborative crafting on the performance of the individuals and teams with whom the crafters interact. If this research indicates that the performance of the non-crafters is adversely affected by job crafting activities, other studies are needed that investigate the strategies that might be introduced to lessen the damage. For example, it may be that any negative effects of crafting on others would be reduced if employees negotiated all customization activities with managers ([Hornung, Rousseau, Glaser, Angerer, & Weigl, 2010](#)) who then coordinated the crafting activities that occurred within the work unit to ensure that they were complementary. Or, it may be that the unintended negative consequences of job crafting might be avoided if crafters themselves coordinated their activities with colleagues. Research is needed that evaluates if either of these strategies is effective in reducing the potential adverse effects of crafting on the performance of others in the work unit and if these strategies change the way that the job crafters themselves respond to these crafting activities.

In addition, research is needed that examines the effects of crafting on the well-being of the crafter's colleagues. A promising start in addressing this issue is found in a study by [Tims et al. \(2015\)](#). These authors demonstrated that when employees craft in order to reduce stressful job demands, their colleagues may be required to take on more tasks and higher workloads, resulting in interpersonal conflict in the work unit and subsequent burnout. This research might be extended by focusing on both individual and collaborative crafting and on activities other than reducing stressful demands. It is possible that many crafting activities require substantial energy and divert the crafter's attention from his or her day-to-day activities which then affects the activities of colleagues and have implications for their well-being. More research is now needed on topics such as these that focus on the unintended consequences of crafting on the well-being of others in the organization.

5.2. The effects of jobs on the organizational context

In previous pages, we discussed research that showed that employee jobs can be constrained by several organizational properties including high levels of formalization and centralization ([Pierce & Dunham, 1978](#); [Rousseau, 1978b](#)). Unfortunately, we know little about the strategies and practices that might be implemented in order to buffer employees from such constraints and controlling policies. Research is needed on this topic. For example, studies might examine the managerial practices that promote employee freedom and autonomy without violating formal policies and job descriptions ([Oldham & Hackman, 2010](#)).

Research might also examine the extent to which bottom-up changes in the design of jobs actually alter the formal properties of organizations. Bottom-up changes of two types warrant our attention. First, managers alone, or in consultation with employees, might develop new and better ways of performing the work (e.g., serving clients) and then introduce these changes in a formal program of job restructuring. Alternatively, employees themselves might take the initiative and introduce substantive improvements while engaging in individual or collaborative job crafting. Research might examine whether either of these approaches to job redesign relax or ease the formal constraints that are in place in the organization in order to accommodate the new designs.

5.3. Generational differences and reactions to jobs

Throughout this article we have discussed the fit between jobs and the personal characteristics of jobholders including their

needs, personalities, and skills. But some recent evidence suggests that beyond these personal characteristics, generational differences may shape individuals' values and preferences and, therefore, have an impact on the way they respond to job properties. Generation is often defined as "an identifiable group that shares birth years, age, location and significant life events at critical developmental stages" (Kupperschmidt, 2000). The standard approach has been to assume that there are now four generations: Veterans (born 1925–42), Baby Boomers (1943–60), Generation X (1961–81), and Generation Y (1982–) (Strauss & Howe, 1991).

Across both cross-sectional and longitudinal studies, there is some evidence to suggest differences in values across the generation groups (Parry & Urwin, 2011), although findings based on cross-sectional research should be interpreted with caution since they may confound developmental and generational effects. For example, Cennamo and Gardner (2008) found generational differences for work values involving status and freedom, but not for extrinsic, social and altruism. Younger generations valued status more than the older group, while Generation Y valued autonomy and work-life balance more than Generation X or Baby Boomers. Chen and Choi (2008) showed that Baby Boomers valued altruism and intellectual stimulation more than Generations X or Y. In addition, Generation Y was less concerned about personal growth (e.g., intellectual stimulation, achievement) and more about economic returns. Consistent with these findings, Twenge, Campbell, Hoffman, and Lance (2010) showed that Generation Y valued intrinsic work rewards less than Boomers or Generation X, but valued more highly work-life balance.

In general, the results suggest that individuals who belong to Generation Y may respond less positively to motivational job characteristics than those from earlier generations. Thus, enriching jobs in a program of formal job restructuring may have fewer benefits for Generation Y employees than for individuals from Generation X or Boomers. In addition, it is possible that generation differences have implications for the nature of job crafting activities that are used by individuals and groups. For example, it may be that Generation Y employees are less likely to customize their jobs by increasing challenging job demands and are more likely to engage in activities that make the work less stressful. Research is now needed to test these possibilities and examine the effects of generation differences on employee reactions to formal job redesign and job crafting.

5.4. Transformational leadership and job design

Transformational leadership has long been one of the most popular approaches to understanding leadership effectiveness (Bass, 1985; Judge & Piccolo, 2004). Transformational leadership typically describes four dimensions of leader behavior: *Idealized influence* (the degree to which leaders behave in charismatic ways that cause followers to identify with them), *inspirational motivation* (the degree to which leaders articulate visions that are appealing to followers), *intellectual stimulation* (the degree to which leaders challenge assumptions, take risks, and solicit followers' ideas), and *individualized consideration* (the degree to which leaders attend to followers' needs, act as mentors or coaches, and listen to followers' concerns). Recent meta-analyses suggest that when leaders engage in such behaviors, employees exhibit generally positive behavioral and attitudinal reactions (e.g., satisfaction, motivation, performance) (Judge & Piccolo, 2004).

Several authors have proposed that the characteristics of employees' jobs play an important role in understanding the effects of transformational leadership (e.g., Shamir, House, & Arthur, 1993). For example, Piccolo and Colquitt (2006) suggest that each of the dimensions of transformational leadership can shape employees' perceptions of job characteristics. That is, leaders

who use intellectual stimulation by seeking new perspectives and developing new ways to perform tasks may enhance employee perceptions of variety and autonomy. Leaders who engage in individualized consideration by coaching and teaching may increase employees' perceptions of autonomy and feedback in their jobs. And when leaders engage in idealized influence or inspirational motivation, employees may view their jobs as more significant.

Previous research provides some support for these arguments showing connections between overall measures of a leader's transformational behavior and both summary measures of job characteristics (Nielsen, Randall, Yarker, & Brenner, 2008; Piccolo & Colquitt, 2006; Purvanova et al., 2006) and measures of individual job characteristics (e.g., Astrauskaite et al., 2015). Moreover, several studies (e.g., Astrauskaite et al., 2015; Gillet & Vandenberghe, 2014; Piccolo & Colquitt, 2006; Purvanova et al., 2006; Wang, Ma, & Zhang, 2014) also show that employees' job characteristics either fully or partially mediate the effects of transformational leadership on various employee outcomes (e.g., harassment, performance, OCBs).

Although the research described above has made substantial contributions to the leadership literature, we believe that additional research is needed in this area. For example, future studies might examine the specific transformational leadership dimensions that shape each of the core job characteristics as well as the social and context dimensions of jobs. In addition, research is needed that examines whether transformational leaders encourage or permit both individual and collaborative job crafting. If this research shows that employees are more likely to craft jobs when transformational leaders are present, follow-up studies might then explore the dimensions of transformational leadership that have the strongest effects on these crafting activities. Finally, since most previous studies on transformational leadership and jobs have been cross-sectional in nature (Piccolo & Colquitt, 2006), future research might use longitudinal designs to determine if employees' job characteristics and job crafting activities shape transformational leadership behavior by creating the conditions and opportunities for such leadership to emerge.

In addition to calling for more research on the effects of transformational leadership on employee job characteristics and job crafting activities, research is needed to examine the possibility that employees' jobs play a different role in understanding the effects of transformational leadership—that of a moderator of the transformational leadership-employee response relationship (Podsakoff et al., 1996). Two forms of moderating effects have been proposed in the early literature and both are deserving of more attention. First, Grant (2012) argued that the presence of certain job characteristics can effectively enhance the effects of transformational behavior. Specifically, he suggested that beneficiary contact can strengthen the effects of transformational leadership by enhancing the salience of the transformational leader's prosocial vision. Results of two studies supported this argument showing that a transformational leadership had stronger effects on performance when beneficiary contact was high (Grant, 2012). Similarly, Podsakoff et al. (1996) showed that the relationship between one dimension of transformational leadership (articulating a vision) and employee satisfaction was more positive when employees performed a task that was "intrinsically satisfying."

Other research suggests that job characteristics may play a different role and essentially serve to substitute for the effects of transformational leadership versus enhance the effects. For example, Podsakoff et al. (1996) demonstrated that task feedback was a substitute for the positive relationship between providing an appropriate role model and trust in the leader. Whittington, Goodwin, and Murray (2004) showed that a summary measure of job characteristics partially substituted for the effects of transformational leadership on affective commitment, but played no role in employee performance or OCBs.

Research is now needed to better understand the possible role of various job characteristics in moderating the effects of transformational leadership. For example, research might examine the specific job characteristics that either substitute or strengthen the four key transformational leadership dimensions. Also, research might examine whether different types of individual or collaborative crafting either neutralize or substitute for the effects of the various dimensions of transformational leadership.

5.5. Job design for temporary employees

Much of the research we have reviewed thus far has focused on the effects of jobs on individuals who are employed by their organizations on a permanent and full-time basis. However, recent estimates suggest that temporary employees and contract workers make up about 12% of the U.S. workforce (Rugaber, 2013). Employees are hired on a temporary basis for a variety of reasons, such as to deal with fluctuations in demands, to reduce costs (e.g., benefits), and to enhance workforce knowledge (Boyce, Ryan, Imus, & Morgeson, 2007; Fried et al., 2008). A number of studies suggest that temporary employees often work side-by-side with full-time employees in what might be characterized as “blended” work environments (Davis-Blake, Broschak, & George, 2003; Kraimer, Wayne, Liden, & Sparrowe, 2005).

The likely interactions between permanent and temporary employees in these blended environments raise a number of interesting research questions relevant to the design of jobs. One of these involves the reactions of permanent employees when temporary employees are present. Previous research suggests that permanents often consider temporaries a threat to their own job security especially if the temporaries work on jobs similar to those of the permanents (Fried et al., 2008; Parker, Griffin, Sprigg, & Wall, 2002). It is possible that permanents will have difficulty focusing on the properties of their jobs when experiencing this threat. Research is needed to test this possibility and examine if permanents react differently to the properties of their jobs and to formal job restructuring programs if they work with temporary employees versus with other permanents.

Relatedly, work is needed to investigate whether temporary employees respond differently to their job characteristics when they work with other temporaries or with permanents. It may be that temporaries have difficulty focusing on their work given their insecure positions and that these feelings of insecurity are amplified when they have contact with permanents. Also of interest would be research that examines the job comparisons (Oldham et al., 1982) made by permanent and temporary employees and whether reactions to their own jobs change as a function of these comparisons. For example, since some evidence suggests that temporaries have relatively little autonomy at work (De Cuyper & De Witte, 2006), comparing their jobs to those of other temporaries may result in more positive outcomes than comparisons to those with permanent jobs.

5.6. Job design across cultures

Much of what we know about the design of jobs is based on research that has been conducted in Western nations and cultures (Fried et al., 2008). Research is now needed that investigates whether the effects of job characteristics vary by culture or nation. Although several studies have been conducted on this topic (see Erez, 2010; Fock, Hui, Au, & Bond, 2013; Gelfand, Erez, & Aycan, 2007; Hauff & Richter, 2015; Robert, Probst, Martocchio, Drasgow, & Lawler, 2000; Roe, Zinovieva, Dienes, & Ten Horn, 2000), the results are generally inconsistent. For example, in a study of employees from 49 countries, Huang and Van De Vliert (2003) found that the relations between intrinsic job characteristics (e.g., autonomy) and job

satisfaction were stronger in more individualistic countries, richer countries, countries with better social welfare programs, and countries with a smaller power distance. Robert et al. (2000) reported that empowerment (autonomy) was negatively associated with satisfaction in India, a country high on power distance, but positively related to satisfaction in the USA, a country low on power distance. And Eylon and Au (1999) reported that the performance of individuals from high power distance cultures was lower when they were empowered versus disempowered.

However, other studies examining motivational job characteristics showed generally consistent effects across cultures (Birnbaum et al., 1986; DeVaro et al., 2007). Research is now needed to explain these inconsistent findings. For example, studies might examine the specific cultures in which employees are most likely to respond to particular job characteristics and the outcomes that are most sensitive to these characteristics.

Research is also needed to determine if different categories of job characteristics (e.g., motivational, social or context) are more salient in certain cultures than in others. It may be, for example, that social job characteristics (e.g., interdependence) play a stronger role than motivational characteristics (e.g., autonomy) in shaping employee reactions in collectivist versus individualistic cultures whereas motivational characteristics play a greater role in individualistic cultures. Also, with regard to autonomous teams, research might examine if such teams are more likely to be implemented in certain cultures than in others and if the conditions described earlier that enhance their effectiveness (e.g., distributed-coordinated leadership) have the same effects across cultures.

Finally, research is now needed that examines whether individual or collaborative crafting is used more frequently and effectively in some cultures than in others. A recent study of employees in the USA and the Netherlands suggests that there may be differences in job crafting across cultures (Gordon, Demerouti, Le Blanc, & Bipp, 2015). Research is now needed to examine if collaborative crafting is used more frequently in cultures characterized as collectivistic and if individual crafting is most often used in individualistic cultures. Finally, studies are needed that examine if job crafters and their coworkers in different cultures respond differently to various individual and collaborative crafting activities.

5.7. Job design and new work arrangements

Much of the previous research on job design focused on individuals or teams performing their jobs in stand-alone organizations. While it is the case that many employees continue to work in these conditions, we are presently in the midst of what we believe are fundamental changes in the relationships among people, the work they do, and the organizations for which they do it. Now employees may telecommute from home. Or they might complete their work at a remote base such as an “urban hub” alongside individuals employed by other organizations (Johns & Gratton, 2013).

We propose that these new work arrangements might fundamentally change the characteristics of employees' jobs. When individuals work from home or in a hub, they are likely to have less face-to-face contact with supervisors and colleagues than those who work in stand-alone organizations. Instead of face-to-face contact, individuals working in these alternative arrangements may have more contact with colleagues via new computing technologies (e.g., email, instant messaging, videoconferencing, file sharing) and have face-to-face contact with individuals not employed by the focal organization (e.g., family members or individuals employed by other companies).

What is needed now is research that examines the impact of these changes on employees' job characteristics. For instance, it may be that individuals experience more autonomy working at

home or in a hub than they do in a stand-alone organization. Specifically, the reduced contact the employee has with managers in these alternative arrangements might reduce the amount of monitoring and control at work which, in turn, enhance freedom and autonomy. Also, those who telecommute may have more control over the scheduling, pace and means at work (Gajendran & Harrison, 2007) again resulting in enhanced autonomy. The new work arrangements might have implications for the amount and type of feedback an individual receives from supervisors and coworkers (Gibson, Gibbs, Stanko, Tesluk, & Cohen, 2011). It may be that these individuals are more reluctant to offer performance feedback using computing technologies than in face-to-face circumstances (Kirkman, Rosen, Tesluk, & Gibson, 2006). However, it is also possible that computing technologies allow individuals to obtain feedback from many others who are not employed by their organizations (Oldham & Da Silva, 2015). Feedback from these individuals could compensate for the loss of feedback from managers and coworkers and might provide new insights and perspectives that help the employee to better perform their jobs. It is also possible that the use of computing technologies will result in excessive feedback and cognitive overload (Grant et al., 2010). Research is needed to sort out the type of feedback individuals receive in these various arrangements and the effects of this feedback on their effectiveness and well-being. Finally, research is needed to determine if high levels of certain job characteristics that emerge as a function of new work arrangements (e.g., autonomy) can compensate for low levels of other characteristics that might also emerge in these conditions.

Job crafting might also be affected by alternative arrangements. Since individuals who work at home or in a remote location should be less visible to supervisors, they may have increased opportunities to engage in personal job crafting (Clegg & Spencer, 2007). Collaborative crafting might occur if teams use computing technologies to complete their work and are located in remote locations. Studies are now needed that examine the impact of remote working and the use of computing technologies on the amount and nature of job crafting.

5.8. The effects of jobs on jobholders

Most of the research on the topic of job design has focused on the effects of various job properties on employee responses, including their performance, attendance, job satisfaction and health (see Fried et al., 2008; Humphrey et al., 2007). Fewer studies have examined whether job properties serve to shape the personal characteristics of jobholders (Brousseau, 1983). We believe more research is needed in this area and suggest several possibilities below.

A few early studies that examined the impact of a job's content on the jobholder focused on the effects of several motivational characteristics. For example, Kohn and Schooler (1982) showed that over a 10-year period, self-directed work led to increased employee intellectual flexibility and a stronger preference for self-directed activities. Brousseau (1978) found that over a 6-year period individuals on jobs with high amounts of task identity and task significance developed an increasingly active orientation. Previous work has also shown that increases in an employee's job control result in increases in his or her self-esteem and self-efficacy beliefs (Parker, 1998), proactive personality (Li, Fay, Frese, Harms, & Gao, 2014), and three of the Big 5 personality traits (openness, conscientiousness, agreeableness) (Wu, 2016). A meta-analysis of several studies showed positive relations between measures of autonomy and job complexity and employees' organization based self-esteem (Bowling, Eschleman, Wang, Kirkendall, & Alarcon, 2010). A few early studies also suggest that motivational characteristics can shape employees' knowledge and skills (e.g., Bond &

Flaxman, 2006; Holman et al., 2012; Taris, Kompier, De Lange, Schaufeli, & Schreurs, 2003). For example, Leach, Wall, and Jackson (2003) showed that when machine operators were given the personal responsibility to complete maintenance tasks, they acquired new skills that allowed them to prevent machine breakdowns.

In summary, the early work described above suggests that an employee's personal characteristics and skills are often shaped by the motivational characteristics in their work, including their autonomy and discretion. Although this work has made a valuable contribution to the literature, it has not focused on many of the personal characteristics shown to be relevant in job design research (e.g., employee needs) or on many of the core job characteristics. Therefore, future studies might examine if individuals' growth needs increase over time if they are assigned new jobs high on the motivational characteristics or if they weaken when assigned jobs low on these characteristics (Kulik, Oldham, & Hackman, 1987). Research might also examine if individuals become more conscientious and open after long-term exposure to jobs with considerable challenge and responsibility. And there is need for research that assesses the specific time intervals necessary before these effects are observed (see Hackman, Pearce, & Wolfe, 1978).

In addition to research on the effects of the motivational characteristics, research is needed on the effects of the social job dimensions that are becoming more prominent in organizations (e.g., dealing with others, interdependence, agent feedback). We know of no research that has examined the effects of such characteristics on individuals' need states or personality. Yet it is possible that individuals with relatively low social needs will become more desirous of social relationships when they are assigned jobs with characteristics that demand interaction with others or when they receive support from these others. It is also possible that individuals will become more extraverted or agreeable after long-term exposure to jobs requiring interaction with others.

The social characteristics present in jobs also might alter an individual's development of new social skills—particularly when the job provides the freedom and opportunity to acquire these skills. For example, if a job required dealing with clients on a regular basis, an employee might gradually hone his or her skills in managing interpersonal relationships. These skills may improve the match between the person and his or her present job and also might be transferable to other jobs in which the employee has an interest. Research is now needed to test these possibilities.

The discussion in the paragraphs above describes the possible effects of traditionally designed jobs. Little research focused on the effects of such jobs on jobholders—but even less research has focused on the effects of job crafting activities. Thus, it is not yet clear if employees who craft individually or collaboratively introduce new challenges in their jobs that stretch their needs, skills or talents beyond what is required by the job or if they simply craft to match their current characteristics and skills. Further, it is not clear if individuals' needs and preferences change simply from having an opportunity to tailor their work or if they change in response to substantive alterations in the work itself. Research is now needed to address the short- and long-term effects of job crafting on the job crafter.

The possibility that employees change in response to the content and demands of their jobs, suggests the need for dynamic versus static models of job design as several authors propose (e.g., Clegg & Spencer, 2007; Frese, Garst, & Fay, 2007). From the perspective of person-job fit (Edwards, 1991; Kulik et al., 1987), as employees change in response to job crafting or to manager imposed job redesign, it may be necessary to further adjust a job's properties in order to achieve a good match. In the case of crafting, this suggests that continuous crafting may be necessary, as Berg,

Wrzesniewski, and Dutton (2010) propose. In the case of job restructuring introduced by managers, this suggests that job redesign is best considered a strategy that managers use on a regular basis making adjustments and alterations to jobs to fit employees' newly developed needs and skills. Research is needed to sort out how frequently job changes should be introduced and which strategies are most effective.

6. Conclusion

Job design is a topic that continues to receive substantial attention from researchers and practitioners alike. In this article we have reviewed much of the early and current research and theory on the topic, including discussing work on Motivation-Hygiene Theory (Herzberg, 1966), JCT (Hackman & Oldham, 1976), and the Interdisciplinary Perspective (Campion, 1988). We then outlined a number of problems and shortcomings of that earlier work and reviewed the literature that attempted to address these shortcomings. This review included a discussion of new moderator variables, alternative job characteristics, alternative personal and work outcomes, and alternative mediators to explain the effects of jobs on employee responses. Finally, we proposed what we believe are several exciting new directions for future studies including a discussion of new work on job crafting, more research on cross-cultural job design, and research that focuses on the effects of jobs on employees' personal characteristics and organizational structures. We hope that our ideas will help stimulate fresh thinking about job design and more research on the topic in the years ahead.

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